Research Document on NOVA Buildathon 2026

Contents

Mind Wandering Detection

Processing Datasets from COG-BCI Dataset

The **COG-BCI database** ([1]) is a multi-session, multi-task EEG dataset designed for research on **passive brain-computer interfaces** (pBCI). Each subject performed the same battery of cognitive tasks across three sessions while **63 EEG channels** were recorded at **500 Hz** with a **common-average reference**. For every condition, the dataset also provides trial-level **behavioral logs**, which makes it possible to align neural activity with overt performance.

The local copy follows a BIDS-like layout, organized per subject (`sub-XX`) and session (`ses-S1`–`ses-S3`):

- `eeg/` – raw EEGLAB files (`.set` header + `.fdt` binary data) for each condition;
- `behavioral/` – trial-level behavioral logs (`.mat`) mirroring the same conditions;
- `chanlocs/` – channel names and 3D coordinates.

Each session contains the same task battery: the N-back task (0/1/2-back), the Flanker task, the Multi-Attribute Task Battery (MATB), the Psychomotor Vigilance Task (PVT), and resting-state blocks (eyes open / eyes closed) at the start and end of the session.

Psychomotor Vigilance Task

The **Psychomotor Vigilance Task** (PVT) is a classic **sustained-attention** paradigm. Participants watch a screen and press a button as soon as a stimulus appears. Stimuli are presented at **pseudo-random inter-stimulus intervals** of 2–10 s, so the task cannot be solved by rhythm or anticipation; performance therefore tracks the current level of vigilance.

The primary outcome is the **reaction time** (RT) on each trial. Slow responses and **lapses** (RTs above roughly 500 ms, or outright misses) are the hallmark of reduced vigilance, which is observed under sleepiness, fatigue, and **mind wandering**. When the mind wanders, attention is diverted away from the stimulus, so the participant reacts late or not at all – precisely the lapse signature that the PVT measures. PVT performance is therefore a standard **behavioral proxy** for mind-wandering episodes, and, combined with the simultaneously recorded EEG, it allows searching for the neural correlates of attentional drift. The PVT EEG is stored in **EEGLAB format**: a `.set` header file together with a `.fdt` binary file that holds the raw samples. Inspecting the MATLAB struct of `PVT.set` gives the following picture:

```
EEG =
  filename: 'PVT.set'   datfile: 'PVT.fdt'
  nbchan:  63           trials: 1
  pnts:    306060       srate:  500
  xmin:    0            xmax:   612.118
  data:    [63×306060 single]
  ref:     'common'
  event:   [272×1 struct]
```

- `nbchan: 63`, `srate: 500` – the recording contains **63 EEG channels** sampled at **500 Hz**.
- `trials: 1`, `pnts: 306060`, `xmax: 612.118` – a **single continuous recording** of 306060 sample points, i.e. roughly **612 seconds (10.2 minutes)**, that is **not** segmented into epochs.
- `data: [63×306060 single]` – the raw amplitude values (in microvolts, single-precision floats) laid out as **channels × time points**.
- `ref: 'common'` – the signal has already been **average-referenced**.
- `chanlocs: [1×63 struct]` – the channel locations, matching `chanlocs/get_chanlocs.txt`.

- `event: [272×1 struct]` – event markers for the 90 PVT trials (stimulus onsets and responses) together with session start/end markers; these events make it possible to **segment the continuous signal into trials**.
- `datfile: 'PVT.fdt'` – the companion binary file containing the actual samples.

The behavioral log `PVT.mat` describes the same 90 trials at the performance level:

Field	Value	Meaning
`reaction_times`	`1×90 double`	Reaction time (s) of each of the 90 trials
`error_trial`	`90×1 double`	Binary flag marking erroneous trials (anticipations / misses)
`reaction_time_error_trial`	`1×90 double`	Reaction time for error trials; large negative sentinel values where no valid RT exists
`isi_time`	`90×1 double`	Inter-stimulus interval preceding each trial
`simin` / `simax`	2 / 10	Design bounds of the ISI in seconds
`ntrials`	90	Total number of trials
`triggers`	`5×2 table`	Counts of the trigger codes used in the task

Taken together, `PVT.set` / `PVT.fdt` provide the continuous neural signal while `PVT.mat` provides the per-trial behavioral ground truth; aligning the two (via `event` and `isi_time`) yields trials labeled by reaction time, which can be thresholded into attentive vs. mind-wandering episodes.

References

References

[1] M. F. Hinss, E. S. Jahanpour, B. Somon, L. Pluchon, F. Dehais, and R. N. Roy, *COG-BCI database: A multi-session and multi-task EEG cognitive dataset for passive brain-computer interfaces.* (Jul. 2022). Zenodo. doi: 10.5281/zenodo.6874129.